

Bias Variance and Random Forest

<u>LB</u>	<u>HV</u>	
Fully grown Decision Tree		HBLV
SVM		Linear /
KNN		Logistic
		Regression

Random Forest reduces high variance (overfitting) by averaging multiple deep, low-bias decision trees trained on bootstrap samples, creating a robust model with lower overall error.

By randomly selecting features at each split, it de-correlates trees to reduce variance. Further, while individual tree depth keeps bias low.